Chapter 1 • Real Numbers

Solved Exercise 1.1

1. Identify each number as rational or irrational.

(i) 2.353535 is a terminating decimal: $2.353535=\frac{2353535}{1000000}$, a ratio of integers.

rational

(ii) Let $x=0.666\ldots$; then $10x=6.666\ldots$ and $9x=6$, so $x=\frac{2}{3}$

rational

(iii) 2.236067... is given as a non-terminating, non-recurring decimal, so it is irrational.

(iv) $\sqrt{7}$ is irrational because 7 is not a perfect square.

irrational

(v) e is a standard non-terminating, non-recurring constant, so it is irrational.

(vi) π is a standard non-terminating, non-recurring constant, so it is irrational.

(vii) If $5+\sqrt{11}$ were rational, subtracting the rational number 5 would make $\sqrt{11}$ rational—a contradiction.

irrational

(viii) Suppose $\sqrt{3}+\sqrt{13}=r$ were rational. Then $r^2=16+2\sqrt{39}$, which would make $\sqrt{39}=\frac{r^2-16}{2}$ rational. This is impossible because 39 is not a perfect square.

irrational

(ix) $\frac{15}{4}$ is already a ratio of integers with non-zero denominator.

rational

(x) $(2-\sqrt{2})(2+\sqrt{2})=2^2-(\sqrt{2})^2=4-2=2$, which is rational.

2(i–ii). Represent √2 and √3 on a number line.

For $\sqrt{2}$, construct a right triangle with perpendicular sides 1 and 1: hypotenuse = $\sqrt{(1^2+1^2)}=\sqrt{2}$.

hypotenuse = $\sqrt{((\sqrt{2})^2+1^2)}=\sqrt{3}$; transfer this length to the number line.

See Figure 2, panels (i) and (ii).

2(iii–vi). Represent 4⅓, −2⅐, $\frac{5}{8}$ and 2¾ on a number line.

4⅓: divide [4,5] into 3 equal parts and mark the first point after 4.

−2⅐: divide [−3,−2] into 7 equal parts and mark the first point left of −2.

$\frac{5}{8}$: divide [0,1] into 8 equal parts and mark the fifth point after 0.

2¾: divide [2,3] into 4 equal parts and mark the third point after 2.

See Figure 2, panels (iii)–(vi).

3(i). Express 0.444… in the form $\frac{p}{q}$

Let x = 0.444...; then 10x = 4.444...

10x − x = 4.444... − 0.444...

⇒ 9x = 4

x = $\frac{4}{9}$

3(ii). Express 0.373737… in the form $\frac{p}{q}$

Let x = 0.373737...; then 100x = 37.373737...

99x = 37

x = $\frac{37}{99}$

3(iii). Express 0.212121… in the form $\frac{p}{q}$

Let x = 0.212121...; then 100x = 21.212121...

99x = 21

x = $\frac{21}{99}$ = $\frac{7}{33}$

4. Name each property used.

(i) (a+4)+b=a+(4+b): the grouping of addends changes.

associative property of addition

(ii) √2+√3=√3+√2: the order of addends changes.

studysapiens.com

commutative property of addition
(iii) $x-x=0$: a number is added to its additive inverse.
(iv) $a(b+c)=ab+ac$: multiplication is distributed over addition.
distributive property
(v) $16+0=16$: adding zero leaves the number unchanged.
additive identity
(vi) $100\times1=100$: multiplying by one leaves the number unchanged.
multiplicative identity
(vii) $4\times(5\times8)=(4\times5)\times8$: the grouping of factors changes.
associative property of multiplication
(viii) $ab=ba$: the order of factors changes.
commutative property of multiplication.

5. Name each order property used.

(i) $-3<-1 \Rightarrow 0<2$: add 3 to both sides; order is preserved.
additive order property
(ii) If a $\frac{1}{b}$: taking reciprocals reverses the order when a,b have the same sign (the required domain assumption).
reciprocal order property for same-signed numbers
(iii) If $a<b$ then $a+c<b+c$: the same number c is added to both sides.
additive order property
(iv) If ac 0 then $a<b$: divide by positive c; order is preserved.
multiplicative order property with c>0
(v) If ac b: divide by negative c; order reverses.
multiplicative order property with c<0 and reversal
(vi) Either $a>b$ or $a=b$ or $a<b$: exactly one of the three relations holds.
trichotomy property.

6. Find two rational numbers between each given pair.

if $a<b$, then $a<\frac{a+b}{2}<b$.

(i) For $\frac{1}{4}$ and $\frac{1}{3}$: first average$=\frac{\frac{1}{4}\frac{+1}{3}}{2}=\frac{7}{24}$; second$=\frac{\frac{1}{4}\frac{+7}{24}}{2}=\frac{13}{48}$
$\frac{7}{24}$ and $\frac{13}{48}$

(ii) For 3 and 4: first average$=\frac{3+4}{2}=\frac{7}{2}$; second$=\frac{3+\frac{7}{2}}{2}=\frac{13}{4}$
$\frac{7}{2}$ and $\frac{13}{4}$

(iii) For $\frac{3}{5}$ and $\frac{4}{5}$: first average$=\frac{\frac{3}{5}\frac{+4}{5}}{2}=\frac{7}{10}$; second$=\frac{\frac{3}{5}\frac{+7}{10}}{2}=\frac{13}{20}$
$\frac{7}{10}$ and $\frac{13}{20}$ Other correct pairs are possible.

studysapiens.com